

Table 1: Master Capability Transition Framework: Phase States, Gate Conditions, and SSL Bands across Four Analytical Dimensions. Gate condition rows state named, verifiable triggers for each phase transition derived from the Critical Enabler cells of the per-priority progression matrices. SSL bands track Strategic Sovereignty Level per phase. KPI anchors serve as cross-reference consistency anchors only.

Phase / SSL Band	R&I Maturity TRL stage · research instruments · innovation outputs	Policy, Regulatory & Standards enforcement · certification · standardisation bodies	Operational Deployment & Adoption pilots · integrations · scale · actor categories	Market Uptake & Strategic Impact market share · SME access · competitiveness · sovereignty
I. Current Status <i>2025 baseline</i> SSL: High Foreign De	Fragmented, academically dominated EU R&I at TRL 3–5 for sovereign alternatives. Heavy dependency on NIST PQC process, Intel TDX/AMD SEV-SNP TEE architectures, and US/Chinese foundation models for SecOps. No coordinated EU-wide applied R&I programme for sovereign secure silicon or edge-native confidential computing. <i>Baseline: sovereign secure silicon TRL ≤4; no EU-manufactured EUCC ‘High’ secure element in production.</i>	Regulatory stack adopted, not yet enforced NIS2 enforcement initiated (Oct. 2024); CRA implementing technical acts pending; AI Act high-risk provisions not yet operational; EUCS non-binding; ENISA PQC guidance absent; NIS2 transposition fragmented across Member States. <i>Baseline: no binding EUCS; no published EU PQC migration guidance; CRA technical acts not yet in force.</i>	No continental sovereign security infrastructure TEE deployment limited to foreign hyperscaler implementations; PQC confined to experimental pilots; Simpl pre-launch; federated SOC at bilateral PoC stage only; Island Mode not standardised; supply chain attestation manual in Open RAN ecosystems. <i>Baseline: 0 Common Data Spaces with Simpl security MIMs; bilateral SOC PoCs only.</i>	Near-zero EU share in critical security components EU domestic production of critical secure silicon near 0%; US/Asian hyperscalers dominate cloud security tooling; no EU-sovereign LLM for SecOps; SME CRA/NIS2 compliance costs unaddressed; RISC-V security products absent from public procurement frameworks. <i>Baseline: EU secure silicon critical infrastructure share <5%; no EU-sovereign LLM for SecOps in deployment.</i>
<i>Gate I → II</i>	R&I: Horizon Europe calls for sovereign secure silicon and lightweight PQC issued; ECCC work programmes activated. Policy: CRA implementing technical acts in force; NIS2 Art. 21 national guidance published by competent authorities. Operational: At least one national pilot per priority domain initiated under regulatory pull. Market: ECCC-co-funded SME compliance support scheme operational; IPCEI-CIS initial deployments contracted.			
II. Phase 1 <i>2025–2030</i> <i>Pilot Validation & Applied Maturity</i> SSL: High → Emergin	Applied R&D (P3,P6,P8,P9) · Fragmented Piloting (P1,P2,P4,P5,P7,P10) Horizon Europe calls targeting sovereign secure silicon, lightweight PQC for 8/16-bit MCUs, Simpl security MIMs, and LLMSecOps. RISC-V secure element prototypes targeting TRL 6–7 under EU Chips Act co-funding. RF fingerprinting and CSI-based device authentication validated on EU-manufactured constrained hardware. ECCC-funded crypto-agility middleware research initiated per FIPS 203/204/205 adoption. <i>KPI: RISC-V prototypes at TRL 6–7; lightweight PQC benchmarked on ARM Cortex-M4 class hardware.</i>	Regulatory activation creates compliance market pull CRA hardware security implementing rules published; NIS2 Art. 21 guidance issued; ENISA PQC sector-specific migration guidelines by 2026, mapping FIPS 203/204/205 to NIS2 obligations; EU AI Act Art. 9 compliance frameworks operational for high-risk security AI; EUCS certification rollout mandating TEE workload protection; Simpl-Open launched with initial security MIM definitions; SBOM templates mandated in EU-funded deployments from 2026. <i>KPI: ENISA PQC guidelines published and adopted by 5+ Member States by 2026.</i>	Pilot deployments under regulatory pull across priority domains 3–5 RISC-V secure element pilots under Horizon Europe; LLM-assisted SOC pilots in 5+ national cybersecurity centres under AI Act Art. 9; CAMARA API profiles in 3+ cross-border 5G corridor pilots by 2027; federated SIEMs with MITRE ATT&CK integration; Simpl security pilots in 2+ Common Data Spaces (Manufacturing, Health); Island Mode fallback tested in NIS2/DORA-compliant edge architectures. <i>KPI: 2+ Common Data Spaces piloting Simpl security MIMs; federated SIEMs active in 3+ Member States.</i>	Sovereignty trajectory established; SME access initiated SME compliance support scheme co-funded by ECCC and national agencies; IPCEI-CIS initial deployments activating bilateral operator interconnection; EU secure silicon below 10% but industrial investment trajectory established under CRA pull; EUCC ‘High’ referenced in public procurement frameworks for critical infrastructure ICT. <i>KPI: EU secure silicon <10% but investment trajectory established; SME support scheme operational in 15+ Member States.</i>

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Phase / SSL Band	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Impact
<i>Gate II → III</i>	R&I: RISC-V prototypes validated at TRL 6–7; lightweight PQC benchmarked on EU-manufactured constrained hardware. Policy: ENISA PQC guidelines published; EUCS rollout active; Simpl-Open launched with security MIM definitions. Operational: Federated SIEM pilots with standardised CTI sharing active in 3+ Member States; 2+ Common Data Space Simpl pilots validated. Market: CRA enforcement generating measurable industrial investment in sovereign hardware alternatives.			
III. Phase 2 2030–2035 Industrial Scaling, Standardisation & Cross-Sector Deployment SSL: Emerging EU →	Applied & industrial R&D concurrent with standardisation Phase 2 is not purely standardisation. Hardware–software co-design for neuromorphic/PIM architectures (P3); sovereign EU LLM/SLM for SecOps at AI Act high-risk compliance (P6); RISC-V TEE designs entering volume production via Chips JU by 2032 (P1, P8); FHE performance optimisation programmes initiated (P8); ZKP-based supply chain attestation R&D (P5); Cognitive Twin prototype development (P9). <i>KPI: At least one RISC-V TEE design in volume production by 2032; sovereign EU LLM/SLM certified at AI Act high-risk level.</i>	Standardisation milestones convert pilots into binding frameworks ETSI TC CYBER hybrid PQC profiles published by 2031 (ML-KEM/FIPS 203; ML-DSA/FIPS 204); CEN/CENELEC JTC 13 Simpl security MIMs ratified by 2032; 3GPP SA3/ETSI TC ITS Island Mode fallback APIs standardised by 2031; Runtime SBOM mandate for critical infrastructure supply chains by 2031; classical-only cryptography deprecated for high-security EU government contexts by 2033; Data Visa legal instruments piloted. <i>KPI: CEN/CENELEC Simpl MIMs ratified; ETSI hybrid PQC profiles published; classical-only deprecated in sovereign contexts by 2033.</i>	Continental-scale federated deployment across regulated sectors First EU RISC-V secure element at EUCC ‘High’ by 2031–2032; hybrid PQC deployed across Industrial IoT, energy grids, and financial systems; Pan-European CTI platform operational as LLMSecOps engine by 2032; UEBA baselines harmonised using STIX/TAXII and MITRE ATT&CK; multi-party privacy-preserving AI pipelines demonstrated in health and finance by 2033; AI-driven graceful degradation standardised in O-RAN AI/ML framework. <i>KPI: 70%+ Common Data Spaces on Simpl-based security; APT mean dwell time reduced by 70% relative to Phase 1 baseline.</i>	EU industrial sovereignty consolidating; 30–40% critical component share EU Chips Act fabrication producing RISC-V TEE designs in volume; EU secure silicon at 30–40% market share in critical infrastructure procurement; sovereign European edge LLM/SLM in operational SecOps deployment; IPCEI-CIS outcomes translated into binding inter-operator SLA frameworks with security audit requirements; Data Visa pilots in Nordic–Baltic and BeNeLux corridors. <i>KPI: EU secure silicon 30–40% critical infrastructure share; 70%+ Data Spaces on Simpl security interoperability.</i>
<i>Gate III → IV</i>	R&I: First EU RISC-V TEE design at EUCC ‘High’ and in volume production; sovereign EU edge LLM/SLM at AI Act high-risk compliance. Policy: CEN/CENELEC Simpl security MIMs ratified; ETSI hybrid PQC profiles published; 3GPP Island Mode APIs standardised. Operational: Pan-European CTI platform operational as LLMSecOps engine; EU secure silicon exceeding 30% critical infrastructure share. Market: EU NaaS consortium offers commercially competitive in 2+ cross-border corridors.			
IV. Phase 3 2035–2040 Autonomous Leadership & Frontier Innovation SSL: Full Sovereign A	Frontier research defining global state of the art FHE reaching operational edge latency thresholds; zero-energy air interface security with ambient RF harvesting; self-evolving AI swarm protocols for autonomous MTD; Cognitive Twin E2E orchestration for cyber-physical resilience; bio-inspired elasticity protocols standardised via ETSI ENI; battery-less node quantum-resistant authentication. EuroHPC hosts federated training infrastructure for self-improving sovereign security AI, preventing regression to non-EU foundation model dependency. <i>KPI: FHE latency within 2× AES-GCM baseline for edge analytics; EuroHPC sovereign security AI model in operational deployment.</i>	EU transitions from standard-taker to global standard-setter EU Cloud Rulebook binding with machine-readable compliance APIs by 2037–2038; EUCC ‘High’ as default procurement requirement for all new EU ICT products; PQC-by-default mandated for all new EU ICT products by 2036; AI Act explainability requirements enforced for autonomous MTD decisions; “Schengen for Cloud” legal framework enacted; ETSI/CEN/CENELEC PQC standards promoted to ISO/ITU-T as global reference implementations. <i>KPI: EU Cloud Rulebook binding by 2038; PQC-by-default mandate in force by 2036; Schengen for Cloud enacted.</i>	Fully autonomous, continent-wide sovereign security operations EU Chips Act sovereign fabrication meeting 80%+ of European secure silicon demand by 2039; FHE operationalised for real-time edge analytics; end-to-end confidential continuum from battery-less IoT to HPC; Cognitive Twins autonomously coordinating E2E cyber-physical recovery; AI-driven MTD continuously shuffling network resources without human intervention; SMPC standardised for cross-border federated AI in regulated sectors (health, finance, defence, justice). <i>KPI: EU secure silicon self-sufficiency ≥80% by 2039; autonomous MTD and Cognitive Twin operational continent-wide.</i>	Competitive parity achieved; standards exported globally European operators achieving competitive scale parity with US/Chinese hyperscalers in cloud-edge NaaS; EU-designed security components dominant in European market; EUCS/EUCC adopted as global reference certification frameworks; “Schengen for Cloud” eliminating national fragmentation; EU cybersecurity technology export balance turning positive; SME participation economically viable without discriminatory compliance barriers. <i>KPI: EU NaaS competitive parity in 3+ cross-border corridors; EUCS/EUCC referenced in ISO/ITU-T standards.</i>

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Phase / SSL Band	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Impact
Gate IV → V	R&I: FHE at operational edge latency; zero-energy authentication validated on battery-less EU hardware; AI swarm protocols stable and AI Act-explainable. Policy: EU Cloud Rulebook binding; Schengen for Cloud enacted; PQC-by-default mandate in force; EU standards submitted to ISO/ITU-T. Operational: EU secure silicon self-sufficiency ≥80%; autonomous MTD and Cognitive Twin E2E orchestration operational continent-wide. Market: EU NaaS competitive parity demonstrated; technology export balance positive.			
V. Strategic End-State Objective 2040 Vision SSL: Full Autonomy	Europe leads the global security R&I agenda EU-originated R&I outputs — PQC, TEE architectures, LLMSecOps, zero-energy security, Cognitive Twins — define the global research agenda and are adopted internationally. Aligned with SNS JU SRIA Phase 3 outcomes: 6G security primitives sovereign and globally deployed. ECCC Strategic Agenda fully operationalised; European industrial cybersecurity competitiveness self-sustaining without emergency public subsidy. <i>Triple accountability: SNS JU SRIA ∩ ECCC Strategic Agenda ∩ EU regulatory stack.</i>	Complete, coherent, machine-enforceable EU regulatory stack NIS2, CRA, AI Act, DORA, Data Act, Cloud Rulebook, and “Schengen for Cloud” operating as a coherent system with machine-readable compliance APIs and no manual audit requirement. EUCS and EUCC recognised globally as reference certification frameworks. ETSI and CEN/CENELEC leading ISO/ITU-T security standardisation tracks. <i>End-state: no compliance gap between deployed sovereign capability and legal obligation; no manual audit required.</i>	Security-by-Design is the default operational baseline Autonomous, self-healing, continent-wide security infrastructure requiring no manual intervention for routine threat response. Full confidential computing continuum from IoT sensor to HPC node. Zero critical operational dependencies on non-EU security infrastructure. Island Mode and graceful degradation are normative operating assumptions, not emergency exceptions. <i>End-state: zero critical non-EU security dependencies; Security-by-Design is default, not option.</i>	Europe: sovereign, competitive, and globally influential Europe is the global reference for sovereign, trustworthy, and sustainable computing security. EU-designed components dominant in European market and competitive globally. SME participation fully economically viable without discriminatory compliance costs. European security standards and AI-driven technologies exported as global reference implementations. Strategic autonomy sustained without emergency policy intervention. <i>End-state: EU cybersecurity technology export balance positive; SME compliance cost non-discriminatory across all 27 Member States.</i>